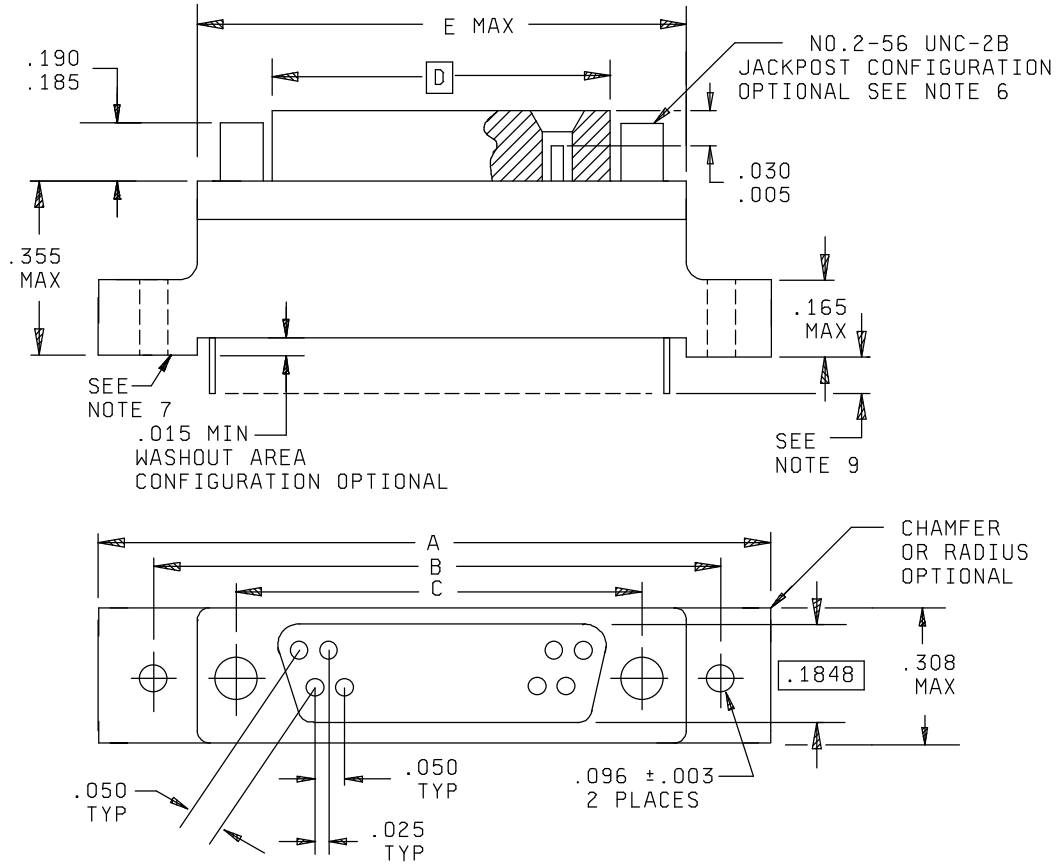
INCH-POUND
MIL-PRF-83513/22B
15 August 1997
SUPERSEDING
MIL-C-83513/22A
14 October 1993

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, STRAIGHT, PIN CONTACTS, 2 ROW ,SOLDER TYPE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83513.

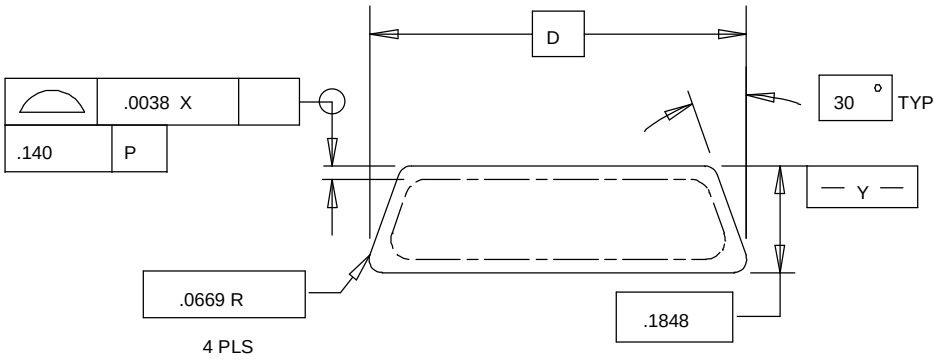


Inches	mm
.003	0.08
.0038	0.097
.015	0.38
.025	0.64
.050	1.27
.0669	1.699
.096	2.44
.140	3.56
.165	4.19
.1848	4.694
.185	4.70
.190	4.83
.308	7.82
.355	9.02

FIGURE 1. Connector, plug, .050 spacing.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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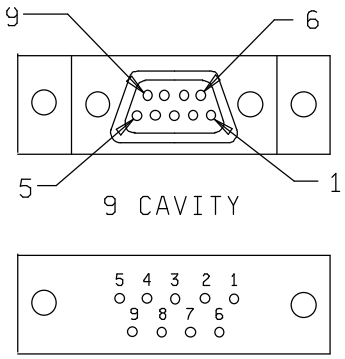
DETAIL OF INTERFACE

Number of contacts	A Max	B ±.007 (0.18 mm)	C ±.005 (0.13)	D	E Max
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.3338 (8.478)	.885 (2.48)
15	1.390 (31.31)	1.150 (29.21)	.715 (18.16)	.4838 (12.289)	.945 (24.00)
21	1.690 (40.64)	1.450 (36.83)	.865 (21.97)	.6338 (16.099)	1.185 (30.10)
25	1.740 (44.20)	1.500 (38.10)	.965 (24.51)	.7338 (18.639)	1.275 (32.39)
31	2.040 (51.82)	1.800 (45.72)	1.115 (28.32)	.8838 (22.449)	1.575 (40.01)
37	2.340 (59.44)	2.100 (53.34)	1.265 (32.13)	1.0338 (26.259)	1.875 (47.63)

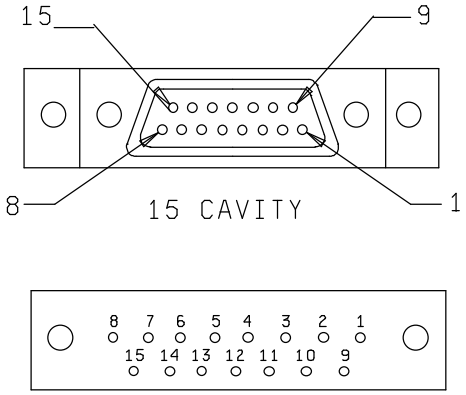
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-PRF-83513. Plastic molding shall conform to the requirements of Diallyl phathalate, or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.
5. Metal shell shall be of material in accordance with MIL-PRF-83513 for class M parts.
6. Jackpost (permanently attached), when specified: Corrosion resistant steel in accordance with ASTM A 484 and ASTM A 582, 300 series stainless steel, passivated in accordance with ASTM A 967.
7. Separately molded plastic body (if used) shall conform to the requirements of Diallyl phathalate, or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.
8. Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.
9. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm) or .172 (4.37 mm). The tolerance shall be ±.015 (0.381 mm) for all termination lengths.

FIGURE 1. Connector, plug, .050 spacing - Continued.

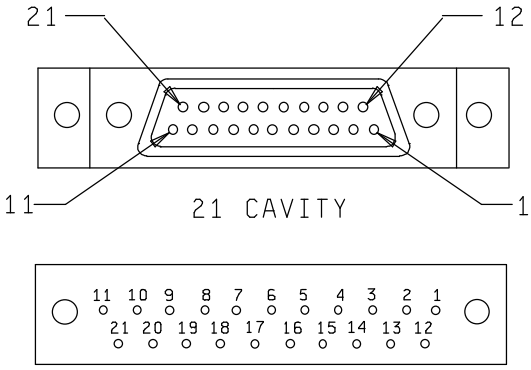
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TERMINATION VIEW - VERTICAL MOUNT



TERMINATION VIEW - VERTICAL MOUNT

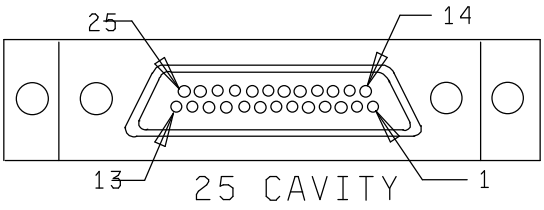


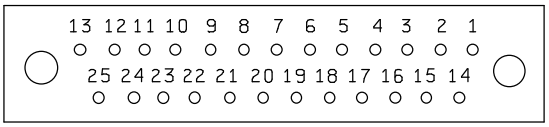
TERMINATION VIEW - VERTICAL MOUNT

NOTE:
1. Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

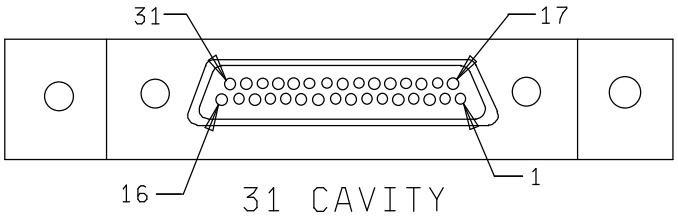
FIGURE 2. Insert arrangement.

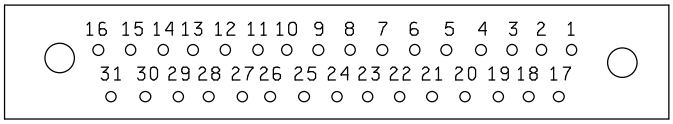
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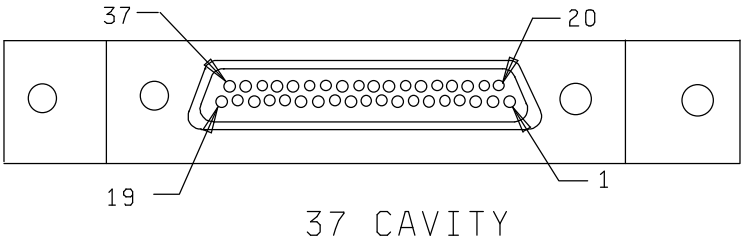


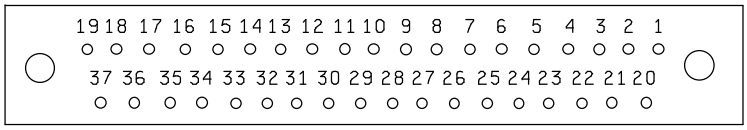
TERMINATION VIEW - VERTICAL MOUNT





TERMINATION VIEW - VERTICAL MOUNT



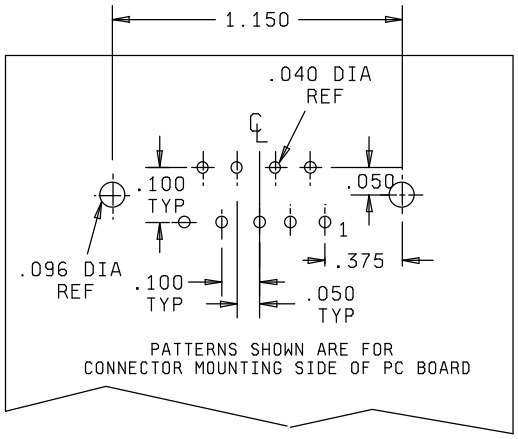


TERMINATION VIEW - VERTICAL MOUNT

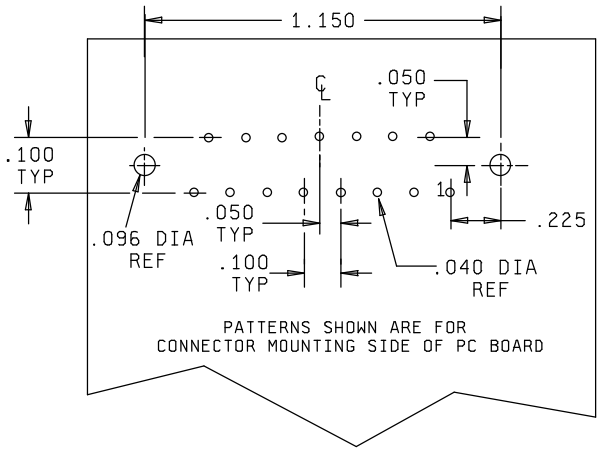
NOTE:
1. Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement - Continued

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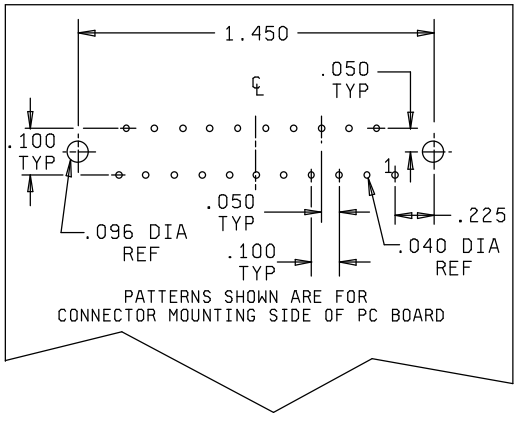


9 CONTACT PLUG



15 CONTACT PLUG

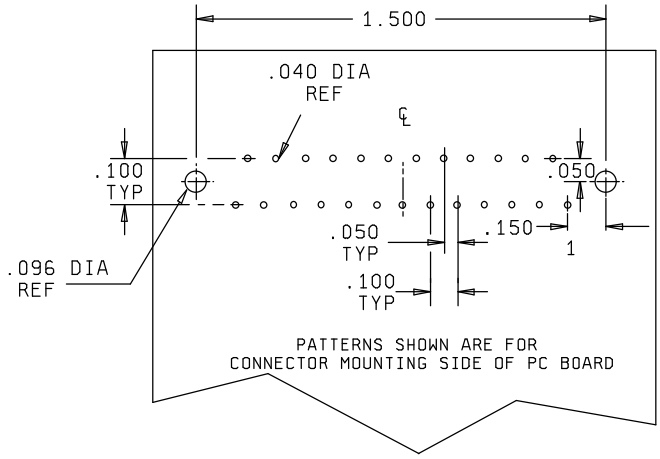
Inches	mm
.040	1.02
.050	1.27
.096	2.44
.100	2.54
.150	3.81
.225	5.72
.375	9.53
1.150	29.21
1.450	36.83
1.500	38.10
1.800	45.72
2.100	53.34



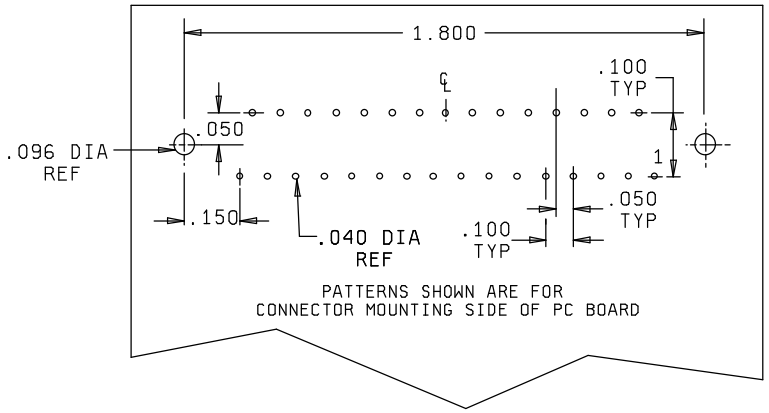
21 CONTACT PLUG

FIGURE 3. Layout arrangement

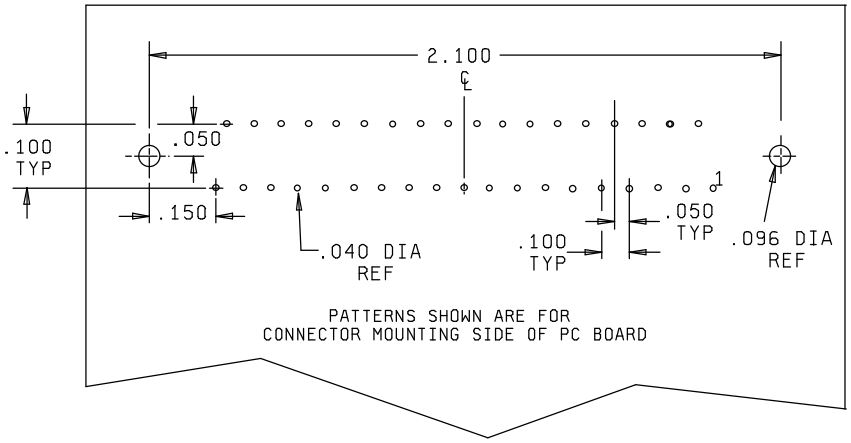
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25 CONTACT PLUG



31 CONTACT PLUG



37 CONTACT PLUG

FIGURE 3. Layout arrangement - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Current rating, maximum: 3 amperes per contact.

Materials:

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-PRF-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-PRF-83513.

Plastic body or plastic molding: Shall conform to the requirements of Diallyl phathalate, or Thermoplastic polyphenylene sulfide or Thermoplastic polyester, glass reinforced liquid crystalline.

Jackpost: Corrosion resistant steel in accordance with ASTM A 484 and ASTM A 582, 300 series stainless steel, passivated in accordance with ASTM A 967.

Wire termination pins: Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.

Mating connector: Shall conform to MIL-PRF-83513/2 and MIL-PRF-83513/4.

Plating of termination leads: Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with J-STD-006.

Part or Identification Number (PIN): The PIN shall consist of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/22-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A= 9 B=15 C=21 D=25 E=31 F=37	Termination length 01=.109 02=.140 03=.172	Shell finish (Interface critical) C = cadmium N = electroless nickel (space application)	Hardware N = no jackpost P = jackpost attached

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - AT, ME, MI
Navy - AS, CG, MC, SH
Air Force - 17, 99

:Preparing activity:
DLA - CC

(Project: 5935-4087-21)

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